

# Extension Guide

---

## Adding a new domain module (backend)

1. Create package under `com.bankone.<module>/` with `controller`, `service`, `repository`, `entity`, `dto` as needed.
2. Wire Spring components (`@RestController`, `@Service`, `@Repository`).
3. Add `SecurityConfig` matchers (do not rely on “authenticated only” for admin mutations).
4. Add entity → verify Hibernate update / add Flyway later.
5. Seed data via `ApplicationRunner` if required.
6. Document: new `docs/MODULES/<Name>.md`, API, schema, call flow, CHANGELOG.

## Adding a new Angular feature

1. Generate/create under `src/app/features/<name>/`.
2. Add route in `app.routes.ts` with `canActivate` + `data.roles`.
3. Add sidebar entry in `core/layout/sidebar/` **only when route is real**.
4. Add `core/services/<name>.ts` + models.
5. Update module docs + CHANGELOG.

## Extending accounts safely

| Goal                   | Recommended approach                                                                                             |
|------------------------|------------------------------------------------------------------------------------------------------------------|
| New account type       | Add <code>AccountType</code> enum value + policy seed + UI label; do not invent numbers without generator update |
| Opening deposit change | Prefer <code>account_policy</code> row over hardcoding                                                           |
| Ledger / withdraw      | Introduce <code>transaction</code> entity; call from deposit/withdraw/open; stop mutating balances alone         |
| Branch master          | New <code>branch</code> table; replace free-text <code>branchCode</code> gradually                               |

## Extending security

- Put roles in matchers early.
- Prefer method security `@PreAuthorize` for fine-grained ops.
- Keep JWT secret out of git for production (deferred hardening).

## Database evolution path

Today: `ddl-auto=update` + `schema.sql` for sequences.

Target: Flyway/Liquibase migrations under `src/main/resources/db/migration`, set `ddl-auto=validate`.

## Stub modules ready for first implementation

| Module      | Empty package                        | Suggested first slice            |
|-------------|--------------------------------------|----------------------------------|
| Transaction | <code>com.bankone.transaction</code> | CREDIT entry on deposit + open   |
| Report      | <code>com.bankone.report</code>      | Daily deposits summary           |
| Audit       | <code>com.bankone.audit</code>       | Persist security/business events |
| Beneficiary | <code>com.bankone.beneficiary</code> | Payee CRUD for transfers         |

## Documentation template for new features

Every new feature doc must include sections 1–20 listed in `docs/README.md` / architect rule, plus `# Future Modification Guide` with files, classes, methods, impact, and call hierarchy.